

Nikita Kazeev

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Research Scientist at the intersection of AI and Physics

Work Experience

National University of Singapore <i>Postdoc under Kostya Novoselov</i>	2022–present
▪ Transformer for generating symmetric crystals : · Created a generative model for material design based on symmetry inductive bias. · Implemented the model in PyTorch. · Achieved the best generation quality and diversity among space-group-conditioned models. · Led a team of 6. ▪ ML for predicting properties of defects in 2D crystals : · Created a sparse representation of crystals with defects, achieving 4x energy prediction error reduction. · Developed code for reproducible parallel running of machine learning experiments. · Led a team of 3.	
Constructor Group <i>Machine Learning Consultant</i>	February 2021–present
▪ AI/ML Consulting : · Consulted on machine learning algorithms, deep learning applications, and architectural designs for scientific and industry-led projects.	
CERN [Yandex → HSE University] <i>Researcher</i> 2025 Breakthrough Prize in Fundamental Physics as a member of LHCb.	2014–2022
▪ Generative models uncertainty estimation : · Developed the first methods for estimating uncertainty of conditional GANs. · Led a team of 3 students. ▪ Machine learning on noisy data : · Developed a rigorous way to train ML algorithms on data with the label-noise model common in high-energy physics. ▪ Muon identification at the LHCb experiment at CERN : · Solved a classification problem over tabular data with noisy labels under timing constraints. · Developed a gradient boosting model that reduced the false positive rate by 30% in the critical low-track-momentum region. · Integrated the model into LHCb production, mostly in C++. · Packaged the work as a data science competition problem for IDAO-2019. ▪ CatBoost aka fighting biases with dynamic boosting : · Set up a distributed system for running experiments for the team with Bash and Python. · Studied gradient boosting improvements with experiments on toy data.	

Education

Higher School of Economics (HSE University) PhD in Computer Science, supervisor Andrey Ustyuzhanin ; Machine Learning for particle identification in the LHCb detector .	2016–2020
Sapienza — Università di Roma PhD in Physics (double degree with HSE), supervisor Barbara Sciascia .	2016–2020
Product Management course at Yandex Focused coursework in product strategy and execution.	Spring 2018

Yandex School of Data Analysis

2013–2015

Master's level CS course covering algorithms, machine learning, deep learning, and distributed systems.

Moscow Institute of Physics and Technology

MS in Physics (2014–2016): Optimisation of data processing of the LHCb experiment.

BS in Physics (2010–2014): Study of the quantum states of the electrons in nonideal plasma and selected molecules using wave packet molecular dynamics with packet splitting.

Mentorship

- Mentored 9 students ([1](#), [2](#), [3](#), [4](#), [5](#), [6](#), [7](#), [8](#), [9](#)), 3 interns, and student workshop projects ([1](#), [20](#)).

Teaching & Outreach

- Taught at the Machine Learning in High Energy Physics Summer School in [2015](#), [2016](#), [2017](#), [2018](#), [2019](#), [2020](#), [2021](#), and [2023](#).
- Lecturer in the [Coursera Advanced Machine Learning specialisation](#), which is [award-winning](#).
- Taught and developed an introductory data analysis [course](#) at HSE.
- Delivered more than 20 [conference talks](#).

Service

- Reviewer for RSC Advances, Machine Learning: Science and Technology, and AI for Accelerated Materials Design workshops at NeurIPS and ICLR (2023–2025).
- Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

Skills

ML & AI (Structured, non-text, domain-specific data) ·
Python (Research coding) ·
PyTorch (Deep learning) ·
C++ (Production) ·
Linux (Administration) ·
Physics (Deep domain understanding)

Selected Publications

1. M. Y. Lukianov, A. Maevskiy, **N. Kazeev**, D. Mylnikov, D. A. Svintsov, K. S. Novoselov, A. Ustyuzhanin, and D. A. Bandurin. [Inverse design of broadband antennas for terahertz devices based on two-dimensional materials](#). *Physical Review Applied* 24, 054079, November 2025.
2. **Kazeev, Nikita**, et al. [WyckoffTransformer: Generation of Symmetric Crystals](#). ICML 2025, May 2025.
3. **Kazeev, N.**, Al-Maeeni, A.R., Romanov, I. et al. [Sparse representation for machine learning the properties of defects in 2D materials](#). *npj Comput Mater* 9, 113 (2023).
4. Anderlini, L., Chimpoesch, C., **Kazeev, N.**, Shishigina, A., & LHCb collaboration. [Generative models uncertainty estimation](#). *Journal of Physics: Conference Series*, 2023. *I'm the corresponding author*.
5. M. Borisyak and **N. Kazeev**. [Machine Learning on data with sPlot background subtraction](#). *Journal of Instrumentation* 14.08 (2019). *I'm the corresponding author*.
6. Derkach, D., **Kazeev, N.**, Ratnikov, F., Ustyuzhanin, A., & Volokhova, A. (2019). [Cherenkov detectors fast simulation using neural networks](#). *Nuclear Instruments and Methods in Physics Research Section A*. *I'm the corresponding author*.